

Raw Data Sample

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Obs	name	MDVP:Fo(Hz)	MDVP:Fhi(Hz)	MDVP:Flo(Hz)	MDVP:Jitter(%)	MDVP:Jitter(Abs)	MDVP:RAP
1	phon_R01_S01_1	119.992	157.302	74.997	0.00784	0.00007	0.0037
2	phon_R01_S01_2	122.4	148.65	113.819	0.00968	0.00008	0.00465
3	phon_R01_S01_3	116.682	131.111	111.555	0.0105	0.00009	0.00544
4	phon_R01_S01_4	116.676	137.871	111.366	0.00997	0.00009	0.00502
5	phon_R01_S01_5	116.014	141.781	110.655	0.01284	0.00011	0.00655
6	phon_R01_S01_6	120.552	131.162	113.787	0.00968	0.00008	0.00463
7	phon_R01_S02_1	120.267	137.244	114.82	0.00333	0.00003	0.00155
8	phon_R01_S02_2	107.332	113.84	104.315	0.0029	0.00003	0.00144
9	phon_R01_S02_3	95.73	132.068	91.754	0.00551	0.00006	0.00293
10	phon_R01_S02_4	95.056	120.103	91.226	0.00532	0.00006	0.00268

Obs	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer(dB)	Shimmer:APQ3	Shimmer:APQ5	MDVP:APQ
1	0.00554	0.01109	0.04374	0.426	0.02182	0.0313	0.02971
2	0.00696	0.01394	0.06134	0.626	0.03134	0.04518	0.04368
3	0.00781	0.01633	0.05233	0.482	0.02757	0.03858	0.0359
4	0.00698	0.01505	0.05492	0.517	0.02924	0.04005	0.03772
5	0.00908	0.01966	0.06425	0.584	0.0349	0.04825	0.04465
6	0.0075	0.01388	0.04701	0.456	0.02328	0.03526	0.03243
7	0.00202	0.00466	0.01608	0.14	0.00779	0.00937	0.01351
8	0.00182	0.00431	0.01567	0.134	0.00829	0.00946	0.01256
9	0.00332	0.0088	0.02093	0.191	0.01073	0.01277	0.01717
10	0.00332	0.00803	0.02838	0.255	0.01441	0.01725	0.02444

Obs	Shimmer:DDA	NHR	HNR	status	RPDE	DFA	spread1	spread2	D2	PPE
1	0.06545	0.02211	21.033	1	0.414783	0.815285	-4.813031	0.266482	2.301442	0.284654
2	0.09403	0.01929	19.085	1	0.458359	0.819521	-4.075192	0.33559	2.486855	0.368674
3	0.0827	0.01309	20.651	1	0.429895	0.825288	-4.443179	0.311173	2.342259	0.332634
4	0.08771	0.01353	20.644	1	0.434969	0.819235	-4.117501	0.334147	2.405554	0.368975
5	0.1047	0.01767	19.649	1	0.417356	0.823484	-3.747787	0.234513	2.33218	0.410335
6	0.06985	0.01222	21.378	1	0.415564	0.825069	-4.242867	0.299111	2.18756	0.357775
7	0.02337	0.00607	24.886	1	0.59604	0.764112	-5.634322	0.257682	1.854785	0.211756
8	0.02487	0.00344	26.892	1	0.63742	0.763262	-6.167603	0.183721	2.064693	0.163755
9	0.03218	0.0107	21.812	1	0.615551	0.773587	-5.498678	0.327769	2.322511	0.231571
10	0.04324	0.01022	21.862	1	0.547037	0.798463	-5.011879	0.325996	2.432792	0.271362

The CONTENTS Procedure

Data Set Name	WORK.PARKINSONS_RENAMED	Observations	195
Member Type	DATA	Variables	24
Engine	V9	Indexes	0
Created	06/08/2026 18:43:28	Observation Length	200
Last Modified	06/08/2026 18:43:28	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	654
Obs in First Data Page	195
Number of Data Set Repairs	0
Filename	/saswork/SAS_workA4B10001BBF1_odaws01-euw1.oda.sas.com/SAS_work19930001BBF1_odaws01-euw1.oda.sas.com/parkinsons_renamed.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	3457684
Access Permission	rw-r--r--
Owner Name	u64477338
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
23	D2	Num	8	BEST12.	BEST32.
20	DFA	Num	8	BEST12.	BEST32.
17	HNR	Num	8	BEST12.	BEST32.
9	Jitter:DDP	Num	8	BEST12.	BEST32.
14	MDVP:APQ	Num	8	BEST12.	BEST32.
6	MDVP:Jitter(Abs)	Num	8	BEST12.	BEST32.
8	MDVP:PPQ	Num	8	BEST12.	BEST32.

The CONTENTS Procedure

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
7	MDVP:RAP	Num	8	BEST12.	BEST32.
11	MDVP:Shimmer(dB)	Num	8	BEST12.	BEST32.
16	NHR	Num	8	BEST12.	BEST32.
24	PPE	Num	8	BEST12.	BEST32.
19	RPDE	Num	8	BEST12.	BEST32.
12	Shimmer:APQ3	Num	8	BEST12.	BEST32.
13	Shimmer:APQ5	Num	8	BEST12.	BEST32.
15	Shimmer:DDA	Num	8	BEST12.	BEST32.
2	avg_fre	Num	8	BEST12.	BEST32.
3	max_fre	Num	8	BEST12.	BEST32.
4	min_fre	Num	8	BEST12.	BEST32.
1	name	Char	14	\$14.	\$14.
21	spread1	Num	8	BEST12.	BEST32.
22	spread2	Num	8	BEST12.	BEST32.
18	status	Num	8	BEST12.	BEST32.
10	var_amp1	Num	8	BEST12.	BEST32.
5	var_fre1	Num	8	BEST12.	BEST32.

The FREQ Procedure

status	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	48	24.62	48	24.62
1	147	75.38	195	100.00

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of avgfre_bin by status			
	avgfre_bin	status		
		0	1	Total
	High	28	24	52
		14.36	12.31	26.67
		53.85	46.15	
		58.33	16.33	
	Low	12	44	56
		6.15	22.56	28.72
		21.43	78.57	
		25.00	29.93	
	Medium	8	79	87
		4.10	40.51	44.62
		9.20	90.80	
		16.67	53.74	
	Total	48	147	195
		24.62	75.38	100.00

WOE Raw Counts

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Obs	avgfre_bin	Good	Bad
1	High	28	24
2	Low	12	44
3	Medium	8	79

WOE Final Table

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Obs	total_good	total_bad	avgfre_bin	Good	Bad	good_dist	bad_dist	WOE
1	48	147	High	28	24	0.58333	0.16327	1.27338
2	48	147	Low	12	44	0.25000	0.29932	-0.18005
3	48	147	Medium	8	79	0.16667	0.53741	-1.17077

WOE Joined Dataset

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Obs	name	avg_fre	max_fre	min_fre	var_fre1	MDVP:Jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DDP
1	phon_R01_S35_5	186.695	211.961	149.442	0.01872	0.0001	0.01075	0.01154	0.03225
2	phon_R01_S10_2	241.404	248.834	232.483	0.00281	0.00001	0.00157	0.00173	0.0047
3	phon_R01_S17_3	222.236	231.345	205.495	0.00266	0.00001	0.00152	0.00144	0.00457
4	phon_R01_S26_3	208.519	220.315	199.02	0.00609	0.00003	0.00368	0.00339	0.01105
5	phon_R01_S34_4	202.805	231.508	86.232	0.0037	0.00002	0.00189	0.00211	0.00568
6	phon_R01_S10_1	237.226	247.326	225.227	0.00298	0.00001	0.00169	0.00182	0.00507
7	phon_R01_S08_5	186.163	197.724	177.584	0.00298	0.00002	0.00165	0.00175	0.00496
8	phon_R01_S50_6	214.289	260.277	77.973	0.00567	0.00003	0.00295	0.00317	0.00885
9	phon_R01_S50_1	201.774	262.707	78.228	0.00694	0.00003	0.00412	0.00396	0.01235
10	phon_R01_S26_2	200.714	223.982	89.686	0.00621	0.00003	0.00371	0.00337	0.01112

Obs	var_amp1	MDVP:Shimmer(dB)	Shimmer:APQ3	Shimmer:APQ5	MDVP:APQ	Shimmer:DDA	NHR	HNR	status
1	0.0583	0.784	0.02699	0.03714	0.06259	0.08096	0.10952	12.529	1
2	0.0176	0.154	0.01006	0.01038	0.01251	0.03017	0.00675	23.145	0
3	0.01643	0.145	0.00867	0.01108	0.012	0.02602	0.0034	25.856	0
4	0.01761	0.155	0.00952	0.01058	0.01307	0.02855	0.0083	22.407	1
5	0.01997	0.18	0.01117	0.01177	0.01506	0.0335	0.0201	18.687	1
6	0.01752	0.164	0.01035	0.01024	0.01133	0.03104	0.0074	22.736	0
7	0.01495	0.135	0.00774	0.00941	0.01233	0.02321	0.00231	26.822	1
8	0.01884	0.19	0.01026	0.01161	0.01373	0.03078	0.04398	21.209	0
9	0.02574	0.255	0.01454	0.01582	0.01758	0.04363	0.04441	19.368	0
10	0.02286	0.207	0.01192	0.01439	0.01852	0.03576	0.00704	23.239	1

Obs	RPDE	DFA	spread1	spread2	D2	PPE	avgfre_bin	WOE
1	0.507826	0.762726	-4.484303	0.340256	2.958815	0.301487	High	1.27338
2	0.457702	0.634267	-6.793547	0.158266	2.256699	0.117399	High	1.27338
3	0.364867	0.694399	-6.966321	0.095882	2.278687	0.103224	High	1.27338
4	0.338097	0.712466	-6.471427	0.184378	2.502336	0.13639	High	1.27338
5	0.536102	0.632631	-5.898673	0.213353	2.470746	0.189032	High	1.27338
6	0.305062	0.654172	-7.31055	0.098648	2.416838	0.095032	High	1.27338
7	0.32648	0.765623	-6.647379	0.201095	2.374073	0.130554	High	1.27338
8	0.462803	0.664357	-5.724056	0.190667	2.555477	0.148569	High	1.27338
9	0.508479	0.683761	-6.934474	0.15989	2.316346	0.112838	High	1.27338
10	0.436084	0.727863	-5.25832	0.256454	2.550961	0.23252	High	1.27338

Obs	_TYPE_	_FREQ_	min1	min2	max1	max2	name	avg_fre	max_fre	min_fre	var_fre1
1	0	195	88.333	102.145	260.105	592.03	phon_R01_S35_5	186.695	211.961	149.442	0.01872
2	0	195	88.333	102.145	260.105	592.03	phon_R01_S10_2	241.404	248.834	232.483	0.00281
3	0	195	88.333	102.145	260.105	592.03	phon_R01_S17_3	222.236	231.345	205.495	0.00266
4	0	195	88.333	102.145	260.105	592.03	phon_R01_S26_3	208.519	220.315	199.02	0.00609
5	0	195	88.333	102.145	260.105	592.03	phon_R01_S34_4	202.805	231.508	86.232	0.0037
6	0	195	88.333	102.145	260.105	592.03	phon_R01_S10_1	237.226	247.326	225.227	0.00298
7	0	195	88.333	102.145	260.105	592.03	phon_R01_S08_5	186.163	197.724	177.584	0.00298
8	0	195	88.333	102.145	260.105	592.03	phon_R01_S50_6	214.289	260.277	77.973	0.00567

Obs	MDVP:Jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DDP	var_amp1	MDVP:Shimmer(dB)	Shimmer:APQ3
1	0.0001	0.01075	0.01154	0.03225	0.0583	0.784	0.02699
2	0.00001	0.00157	0.00173	0.0047	0.0176	0.154	0.01006
3	0.00001	0.00152	0.00144	0.00457	0.01643	0.145	0.00867
4	0.00003	0.00368	0.00339	0.01105	0.01761	0.155	0.00952
5	0.00002	0.00189	0.00211	0.00568	0.01997	0.18	0.01117
6	0.00001	0.00169	0.00182	0.00507	0.01752	0.164	0.01035
7	0.00002	0.00165	0.00175	0.00496	0.01495	0.135	0.00774
8	0.00003	0.00295	0.00317	0.00885	0.01884	0.19	0.01026

Obs	Shimmer:APQ5	MDVP:APQ	Shimmer:DDA	NHR	HNR	status	RPDE	DFA	spread1	spread2
1	0.03714	0.06259	0.08096	0.10952	12.529	1	0.507826	0.762726	-4.484303	0.340256
2	0.01038	0.01251	0.03017	0.00675	23.145	0	0.457702	0.634267	-6.793547	0.158266
3	0.01108	0.012	0.02602	0.0034	25.856	0	0.364867	0.694399	-6.966321	0.095882
4	0.01058	0.01307	0.02855	0.0083	22.407	1	0.338097	0.712466	-6.471427	0.184378
5	0.01177	0.01506	0.0335	0.0201	18.687	1	0.536102	0.632631	-5.898673	0.213353
6	0.01024	0.01133	0.03104	0.0074	22.736	0	0.305062	0.654172	-7.31055	0.098648
7	0.00941	0.01233	0.02321	0.00231	26.822	1	0.32648	0.765623	-6.647379	0.201095
8	0.01161	0.01373	0.03078	0.04398	21.209	0	0.462803	0.664357	-5.724056	0.190667

Obs	D2	PPE	avgfre_bin	WOE	norm_avg	norm_max
1	2.958815	0.301487	High	1.27338	0.57263	0.22417
2	2.256699	0.117399	High	1.27338	0.89113	0.29944
3	2.278687	0.103224	High	1.27338	0.77954	0.26374
4	2.502336	0.13639	High	1.27338	0.69968	0.24122
5	2.470746	0.189032	High	1.27338	0.66642	0.26407
6	2.416838	0.095032	High	1.27338	0.86681	0.29636
7	2.374073	0.130554	High	1.27338	0.56953	0.19510
8	2.555477	0.148569	High	1.27338	0.73327	0.32279

Obs	_TYPE_	_FREQ_	min1	min2	max1	max2	name	avg_fre	max_fre	min_fre	var_fre1
9	0	195	88.333	102.145	260.105	592.03	phon_R01_S50_1	201.774	262.707	78.228	0.00694
10	0	195	88.333	102.145	260.105	592.03	phon_R01_S26_2	200.714	223.982	89.686	0.00621

Obs	MDVP:Jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DDP	var_amp1	MDVP:Shimmer(dB)	Shimmer:APQ3
9	0.00003	0.00412	0.00396	0.01235	0.02574	0.255	0.01454
10	0.00003	0.00371	0.00337	0.01112	0.02286	0.207	0.01192

Obs	Shimmer:APQ5	MDVP:APQ	Shimmer:DDA	NHR	HNR	status	RPDE	DFA	spread1	spread2
9	0.01582	0.01758	0.04363	0.04441	19.368	0	0.508479	0.683761	-6.934474	0.15989
10	0.01439	0.01852	0.03576	0.00704	23.239	1	0.436084	0.727863	-5.25832	0.256454

Obs	D2	PPE	avgfre_bin	WOE	norm_avg	norm_max
9	2.316346	0.112838	High	1.27338	0.66042	0.32775
10	2.550961	0.23252	High	1.27338	0.65425	0.24871

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling
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Input Data Set	PARKINSONS_FINAL
Random Number Seed	123
Sampling Rate	0.8
Sample Size	156
Selection Probability	0.8
Sampling Weight	0
Output Data Set	TRAIN_CV

The FREQ Procedure

status	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	40	25.64	40	25.64
1	116	74.36	156	100.00

The FREQ Procedure

status	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	8	20.51	8	20.51
1	31	79.49	39	100.00

The FREQ Procedure

Fold	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	32	20.51	32	20.51
2	31	19.87	63	40.38
3	31	19.87	94	60.26
4	31	19.87	125	80.13
5	31	19.87	156	100.00

The LOGISTIC Procedure

Model Information	
Data Set	WORK.CV_TRAIN
Response Variable	status
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	124
Number of Observations Used	124

Response Profile		
Ordered Value	status	Total Frequency
1	1	93
2	0	31

Probability modeled is status='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	141.459	132.299
SC	144.279	143.581
-2 Log L	139.459	124.299

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	15.1597	3	0.0017
Score	16.0916	3	0.0011
Wald	14.4653	3	0.0023

The LOGISTIC Procedure

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	1.5159	0.5656	7.1838	0.0074
norm_avg	1	-1.3088	1.3159	0.9892	0.3199
norm_max	1	0.5155	1.4131	0.1331	0.7153
WOE	1	-0.5821	0.2905	4.0168	0.0450

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
norm_avg	0.270	0.020	3.562
norm_max	1.674	0.105	26.711
WOE	0.559	0.316	0.987

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	72.8	Somers' D	0.456
Percent Discordant	27.2	Gamma	0.456
Percent Tied	0.0	Tau-a	0.172
Pairs	2883	c	0.728

The LOGISTIC Procedure

Model Information	
Data Set	WORK.CV_TRAIN
Response Variable	status
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	125
Number of Observations Used	125

Response Profile		
Ordered Value	status	Total Frequency
1	1	94
2	0	31

Probability modeled is status='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	142.032	128.444
SC	144.860	139.757
-2 Log L	140.032	120.444

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	19.5876	3	0.0002
Score	20.6586	3	0.0001
Wald	17.9957	3	0.0004

The LOGISTIC Procedure

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	1.8799	0.5924	10.0696	0.0015
norm_avg	1	-1.2500	1.2201	1.0496	0.3056
norm_max	1	-1.1288	1.0912	1.0700	0.3009
WOE	1	-0.6016	0.2866	4.4060	0.0358

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
norm_avg	0.287	0.026	3.131
norm_max	0.323	0.038	2.745
WOE	0.548	0.312	0.961

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	76.3	Somers' D	0.525
Percent Discordant	23.7	Gamma	0.525
Percent Tied	0.0	Tau-a	0.197
Pairs	2914	c	0.763

The LOGISTIC Procedure

Model Information	
Data Set	WORK.CV_TRAIN
Response Variable	status
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	125
Number of Observations Used	125

Response Profile		
Ordered Value	status	Total Frequency
1	1	95
2	0	30

Probability modeled is status='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	139.770	124.260
SC	142.598	135.573
-2 Log L	137.770	116.260

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	21.5098	3	<.0001
Score	22.8168	3	<.0001
Wald	19.5888	3	0.0002

The LOGISTIC Procedure

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	1.5859	0.5707	7.7216	0.0055
norm_avg	1	-0.9538	1.2864	0.5498	0.4584
norm_max	1	-0.1205	1.3278	0.0082	0.9277
WOE	1	-0.8026	0.3048	6.9342	0.0085

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
norm_avg	0.385	0.031	4.794
norm_max	0.886	0.066	11.965
WOE	0.448	0.247	0.814

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	79.4	Somers' D	0.588
Percent Discordant	20.6	Gamma	0.588
Percent Tied	0.0	Tau-a	0.216
Pairs	2850	c	0.794

The LOGISTIC Procedure

Model Information	
Data Set	WORK.CV_TRAIN
Response Variable	status
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	125
Number of Observations Used	125

Response Profile		
Ordered Value	status	Total Frequency
1	1	92
2	0	33

Probability modeled is status='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	146.300	132.095
SC	149.128	143.408
-2 Log L	144.300	124.095

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	20.2053	3	0.0002
Score	20.8592	3	0.0001
Wald	18.3854	3	0.0004

The LOGISTIC Procedure

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	1.5229	0.5529	7.5870	0.0059
norm_avg	1	-0.3615	1.2410	0.0849	0.7708
norm_max	1	-1.7654	1.2060	2.1429	0.1432
WOE	1	-0.7415	0.2886	6.6033	0.0102

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
norm_avg	0.697	0.061	7.931
norm_max	0.171	0.016	1.819
WOE	0.476	0.271	0.839

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	73.8	Somers' D	0.476
Percent Discordant	26.2	Gamma	0.476
Percent Tied	0.0	Tau-a	0.187
Pairs	3036	c	0.738

The LOGISTIC Procedure

Model Information	
Data Set	WORK.CV_TRAIN
Response Variable	status
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	125
Number of Observations Used	125

Response Profile		
Ordered Value	status	Total Frequency
1	1	90
2	0	35

Probability modeled is status='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	150.238	131.794
SC	153.067	143.107
-2 Log L	148.238	123.794

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	24.4445	3	<.0001
Score	25.4099	3	<.0001
Wald	21.9125	3	<.0001

The LOGISTIC Procedure

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	1.6550	0.5696	8.4424	0.0037
norm_avg	1	-1.3313	1.2014	1.2279	0.2678
norm_max	1	-0.7231	1.0726	0.4546	0.5002
WOE	1	-0.7044	0.2871	6.0173	0.0142

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
norm_avg	0.264	0.025	2.783
norm_max	0.485	0.059	3.971
WOE	0.494	0.282	0.868

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	77.0	Somers' D	0.541
Percent Discordant	23.0	Gamma	0.541
Percent Tied	0.0	Tau-a	0.220
Pairs	3150	c	0.770

The LOGISTIC Procedure

Model Information	
Data Set	WORK.TRAIN_DATA
Response Variable	status
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	156
Number of Observations Used	156

Response Profile		
Ordered Value	status	Total Frequency
1	1	116
2	0	40

Probability modeled is status='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	179.612	161.067
SC	182.662	173.266
-2 Log L	177.612	153.067

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	24.5451	3	<.0001
Score	25.8464	3	<.0001
Wald	22.6817	3	<.0001

The LOGISTIC Procedure

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	1.6263	0.5076	10.2664	0.0014
norm_avg	1	-1.0197	1.1051	0.8514	0.3562
norm_max	1	-0.6782	1.0478	0.4190	0.5174
WOE	1	-0.6862	0.2592	7.0090	0.0081

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
norm_avg	0.361	0.041	3.147
norm_max	0.508	0.065	3.957
WOE	0.503	0.303	0.837

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	75.3	Somers' D	0.505
Percent Discordant	24.7	Gamma	0.505
Percent Tied	0.0	Tau-a	0.194
Pairs	4640	c	0.753

Final Model Coefficients

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Obs	Variable	DF	Estimate	StdErr	WaldChiSq	ProbChiSq	_ESTTYPE_
1	Intercept	1	1.6263	0.5076	10.2664	0.0014	MLE
2	norm_avg	1	-1.0197	1.1051	0.8514	0.3562	MLE
3	norm_max	1	-0.6782	1.0478	0.4190	0.5174	MLE
4	WOE	1	-0.6862	0.2592	7.0090	0.0081	MLE

Test Predictions

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Obs	Selected	_TYPE_	_FREQ_	min1	min2	max1	max2	name	avg_fre	max_fre	min_fre	var_fre1
1	0	0	195	88.333	102.145	260.105	592.03	phon_R01_S26_3	208.519	220.315	199.02	0.00609
2	0	0	195	88.333	102.145	260.105	592.03	phon_R01_S10_1	237.226	247.326	225.227	0.00298
3	0	0	195	88.333	102.145	260.105	592.03	phon_R01_S26_2	200.714	223.982	89.686	0.00621
4	0	0	195	88.333	102.145	260.105	592.03	phon_R01_S07_6	201.464	210.565	195.708	0.00198
5	0	0	195	88.333	102.145	260.105	592.03	phon_R01_S35_6	192.818	224.429	168.793	0.03107
6	0	0	195	88.333	102.145	260.105	592.03	phon_R01_S17_2	223.365	238.987	98.664	0.00264
7	0	0	195	88.333	102.145	260.105	592.03	phon_R01_S17_6	228.969	239.541	113.201	0.00238
8	0	0	195	88.333	102.145	260.105	592.03	phon_R01_S10_5	245.51	262.09	231.848	0.00235

Obs	MDVP:Jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DDP	var_amp1	MDVP:Shimmer(dB)	Shimmer:APQ3	Shimmer:APQ5
1	0.00003	0.00368	0.00339	0.01105	0.01761	0.155	0.00952	0.01058
2	0.00001	0.00169	0.00182	0.00507	0.01752	0.164	0.01035	0.01024
3	0.00003	0.00371	0.00337	0.01112	0.02286	0.207	0.01192	0.01439
4	0.00001	0.00105	0.00115	0.00314	0.01194	0.107	0.00586	0.0076
5	0.00016	0.018	0.01958	0.05401	0.11908	1.302	0.05647	0.0794
6	0.00001	0.00154	0.00151	0.00461	0.01906	0.165	0.01013	0.01296
7	0.00001	0.00136	0.0014	0.00408	0.01745	0.154	0.00942	0.0116
8	0.00001	0.00127	0.00148	0.0038	0.01608	0.141	0.00906	0.00977

Obs	MDVP:APQ	Shimmer:DDA	NHR	HNR	status	RPDE	DFA	spread1	spread2	D2	PPE
1	0.01307	0.02855	0.0083	22.407	1	0.338097	0.712466	-6.471427	0.184378	2.502336	0.13639
2	0.01133	0.03104	0.0074	22.736	0	0.305062	0.654172	-7.31055	0.098648	2.416838	0.095032
3	0.01852	0.03576	0.00704	23.239	1	0.436084	0.727863	-5.25832	0.256454	2.550961	0.23252
4	0.00957	0.01758	0.00135	31.732	0	0.344252	0.742737	-7.777685	0.170183	2.447064	0.05761
5	0.13778	0.16942	0.21713	8.441	1	0.625866	0.76832	-2.434031	0.450493	3.079221	0.527367
6	0.0134	0.03039	0.00301	26.138	0	0.447979	0.686264	-7.293801	0.086372	2.32156	0.098555
7	0.01234	0.02827	0.0042	24.547	0	0.305429	0.681811	-7.314237	0.006274	2.118596	0.091546
8	0.01149	0.02719	0.00476	24.602	0	0.467489	0.631653	-7.156076	0.127642	2.392122	0.097336

Obs	avgfre_bin	WOE	norm_avg	norm_max	F_status	I_status	P_1	P_0
1	High	1.27338	0.69968	0.24122	1	0	0.46889	0.53111
2	High	1.27338	0.86681	0.29636	0	0	0.41766	0.58234
3	High	1.27338	0.65425	0.24871	1	0	0.47918	0.52082
4	High	1.27338	0.65861	0.22132	0	0	0.48271	0.51729
5	High	1.27338	0.60828	0.24962	1	0	0.49073	0.50927
6	High	1.27338	0.78611	0.27933	0	0	0.44064	0.55936
7	High	1.27338	0.81874	0.28047	0	0	0.43227	0.56773
8	High	1.27338	0.91503	0.32649	0	0	0.40083	0.59917

Test Predictions

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Obs	Selected	_TYPE_	_FREQ_	min1	min2	max1	max2	name	avg_fre	max_fre	min_fre	var_fre1
9	0	0	195	88.333	102.145	260.105	592.03	phon_R01_S42_6	244.99	272.21	239.17	0.00451
10	0	0	195	88.333	102.145	260.105	592.03	phon_R01_S16_5	116.879	131.897	108.153	0.00788

Obs	MDVP:Jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DDP	var_amp1	MDVP:Shimmer(dB)	Shimmer:APQ3	Shimmer:APQ5
9	0.00002	0.00279	0.00237	0.00837	0.01897	0.181	0.01084	0.01121
10	0.00007	0.00334	0.00493	0.01003	0.02645	0.265	0.01394	0.01625

Obs	MDVP:APQ	Shimmer:DDA	NHR	HNR	status	RPDE	DFA	spread1	spread2	D2	PPE
9	0.01255	0.03253	0.01049	21.528	0	0.522812	0.646818	-7.3045	0.171088	2.095237	0.09622
10	0.02137	0.04183	0.00786	22.603	1	0.540049	0.813432	-4.476755	0.262633	1.827012	0.326197

Obs	avgfre_bin	WOE	norm_avg	norm_max	F_status	I_status	P_1	P_0
9	High	1.27338	0.91201	0.34715	0	0	0.39821	0.60179
10	Low	-0.18005	0.16619	0.06073	1	1	0.82334	0.17666

The FREQ Procedure

Frequency

Table of status by pred			
status	pred		
	0	1	Total
0	6	2	8
1	3	28	31
Total	9	30	39

Obs	TP	TN	FP	FN	Accuracy	Precision	Recall	F1
1	28	6	2	3	0.87179	0.93333	0.90323	0.91803

The LOGISTIC Procedure

Model Information	
Data Set	WORK.TRAIN_DATA
Response Variable	status
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	156
Number of Observations Used	156

Response Profile		
Ordered Value	status	Total Frequency
1	0	40
2	1	116

Probability modeled is status='0'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

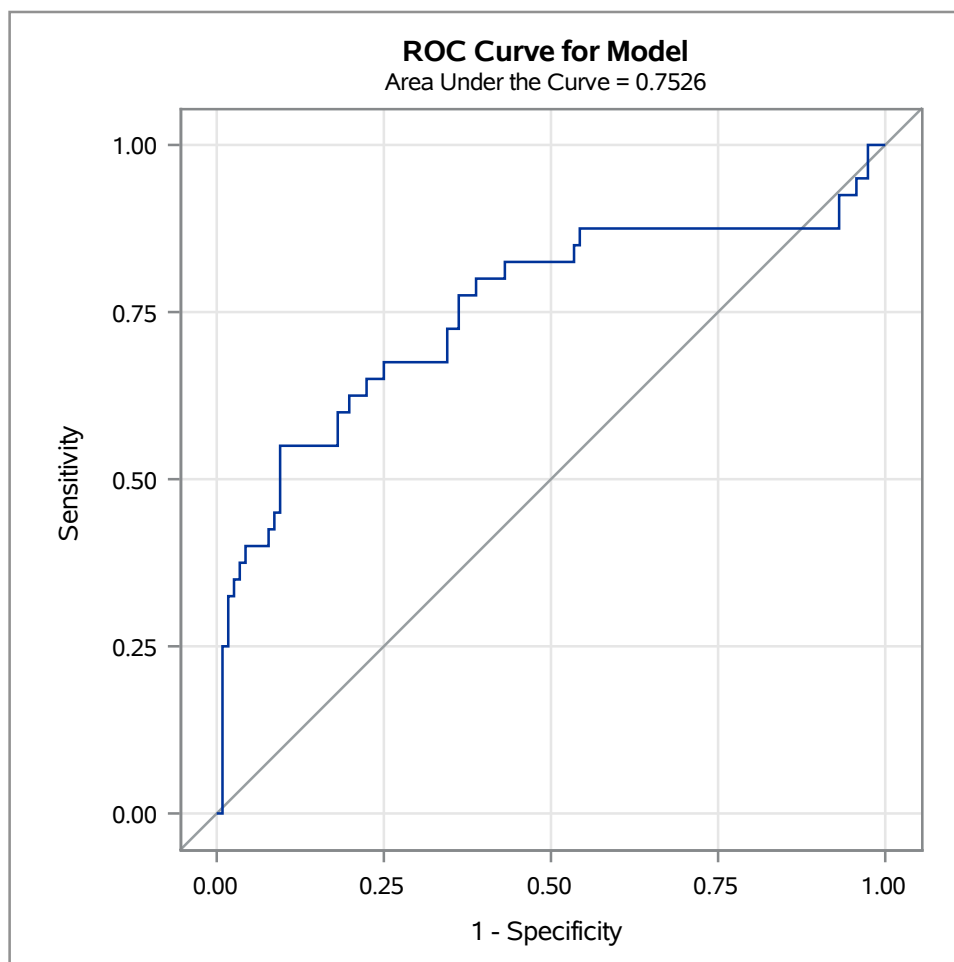
Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	179.612	161.067
SC	182.662	173.266
-2 Log L	177.612	153.067

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	24.5451	3	<.0001
Score	25.8464	3	<.0001
Wald	22.6817	3	<.0001

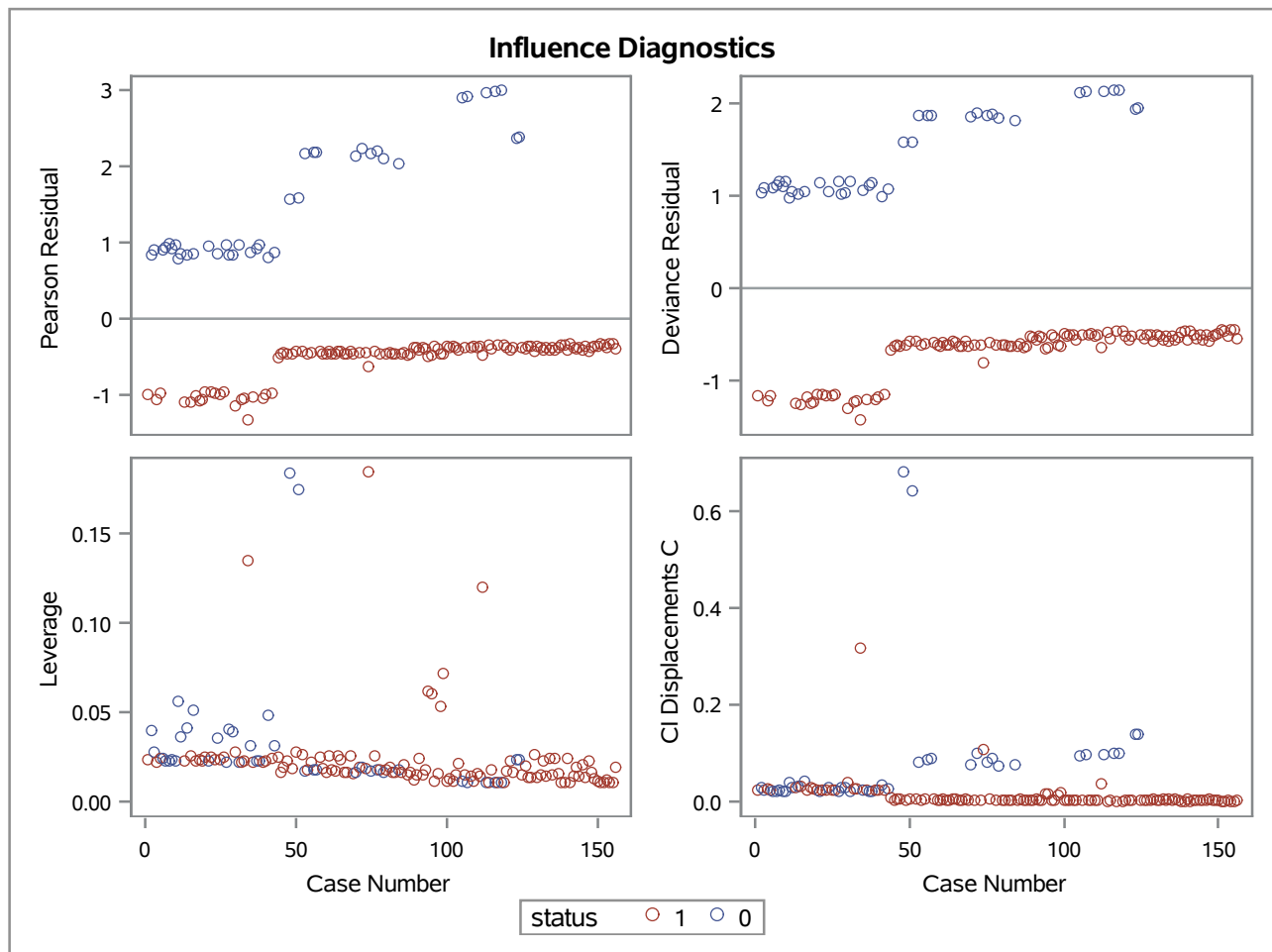
The LOGISTIC Procedure

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-1.6263	0.5076	10.2664	0.0014
norm_avg	1	1.0197	1.1051	0.8514	0.3562
norm_max	1	0.6782	1.0478	0.4190	0.5174
WOE	1	0.6862	0.2592	7.0090	0.0081

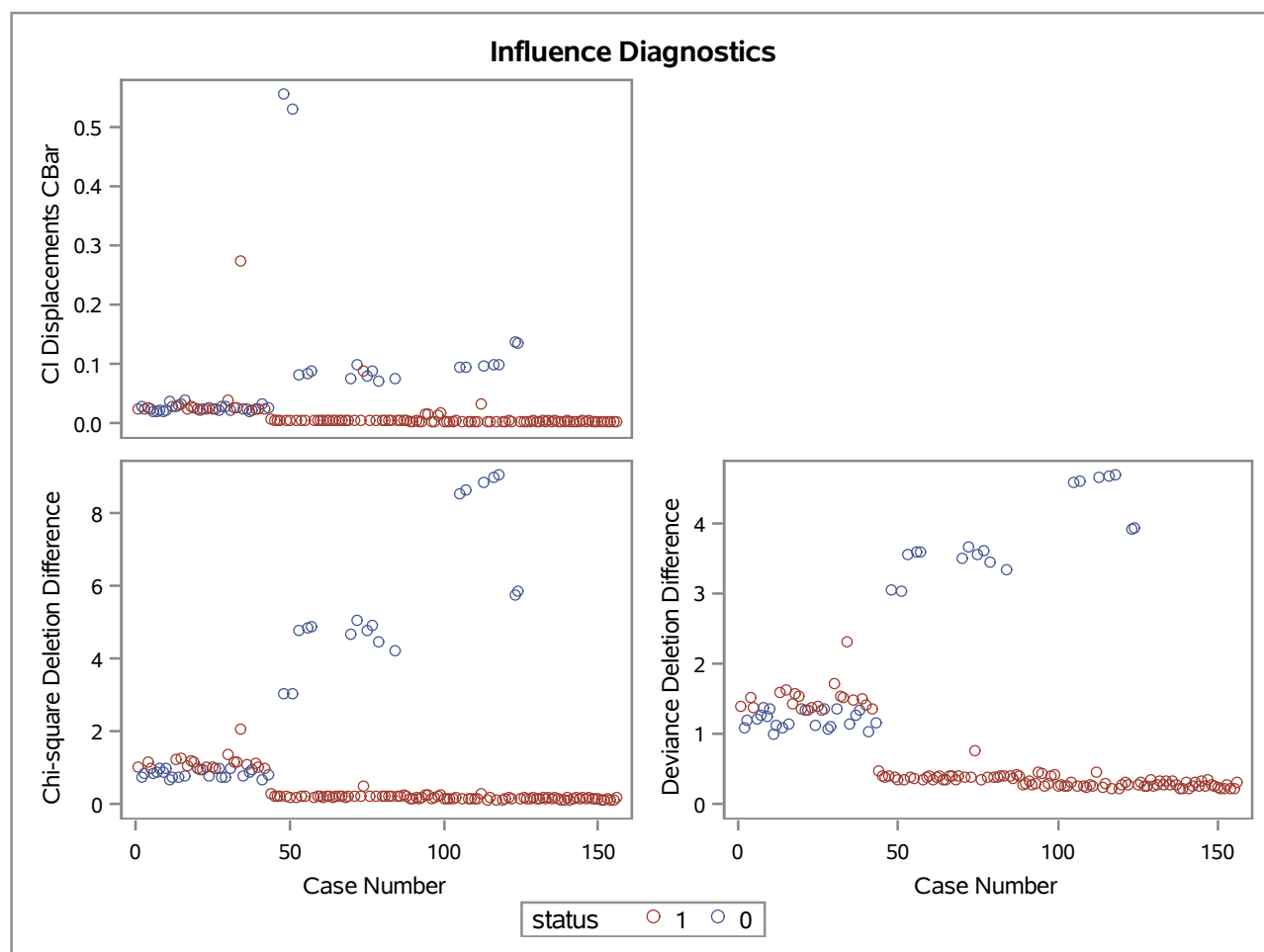
Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
norm_avg	2.772	0.318	24.186
norm_max	1.970	0.253	15.363
WOE	1.986	1.195	3.301



The LOGISTIC Procedure



The LOGISTIC Procedure



ROC Model: ROC1

Model Convergence Status

Convergence criterion (GCONV=1E-8) satisfied.

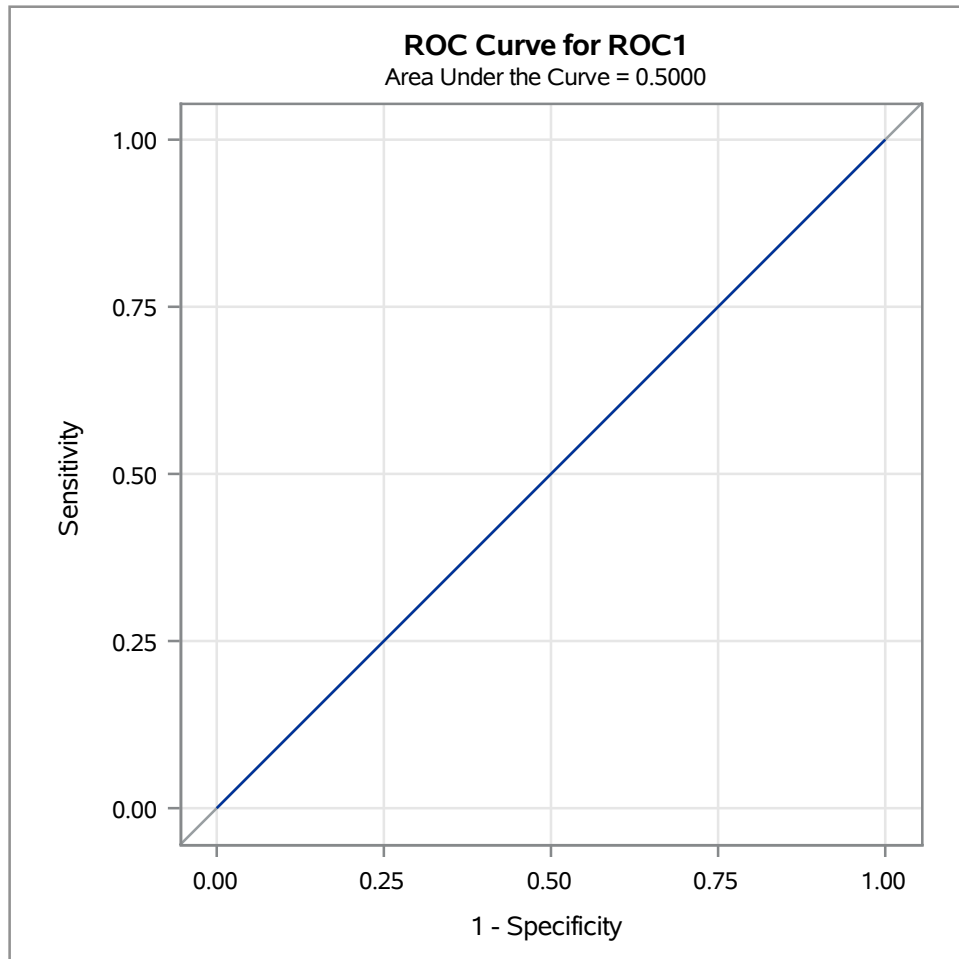
-2 Log L	=	177.612
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Analysis of Maximum Likelihood Estimates

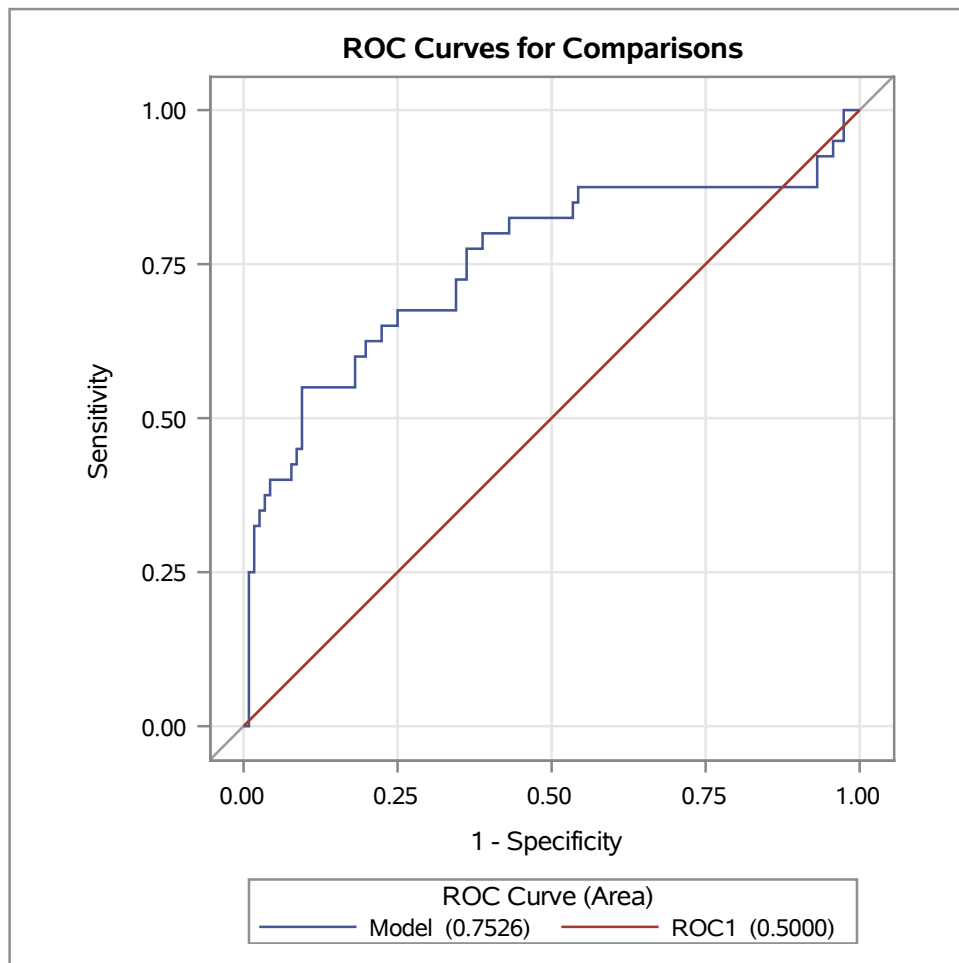
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-1.0647	0.1834	33.7176	<.0001

The LOGISTIC Procedure

ROC Model: ROC1



The LOGISTIC Procedure



ROC Association Statistics							
ROC Model	Mann-Whitney				Somers' D	Gamma	Tau-a
	Area	Standard Error	95% Wald Confidence Limits				
Model	0.7526	0.0520	0.6506	0.8546	0.5052	0.5052	0.1939
ROC1	0.5000	0	0.5000	0.5000	0	.	0

Obs	ROCModel	Area	StdErr	LowerArea	UpperArea	SomersD	Gamma	TauA
1	Model	0.7526	0.0520	0.6506	0.8546	0.5052	0.5052	0.1939
2	ROC1	0.5000	0	0.5000	0.5000	0	.	0

New Patient Prediction

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Obs	_TYPE_	_FREQ_	min1	min2	max1	max2	avg_fre	max_fre	avgfre_bin	WOE
1	0	195	88.333	102.145	260.105	592.03	130	200	Medium	0.2

Obs	norm_avg	norm_max	I_status	P_1	P_0
1	0.24257	0.19975	1	0.75142	0.24858

New Patient Prediction

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Obs	total_good	total_bad	avgfre_bin	Good	Bad	good_dist	bad_dist	WOE
1	48	147	High	28	24	0.58333	0.16327	1.27338
2	48	147	Low	12	44	0.25000	0.29932	-0.18005
3	48	147	Medium	8	79	0.16667	0.53741	-1.17077

New Patient Prediction

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Obs	TP	TN	FP	FN	Accuracy	Precision	Recall	F1
1	28	6	2	3	0.87179	0.93333	0.90323	0.91803

Obs	ROCModel	Area	StdErr	LowerArea	UpperArea	SomersD	Gamma	TauA
1	Model	0.7526	0.0520	0.6506	0.8546	0.5052	0.5052	0.1939
2	ROC1	0.5000	0	0.5000	0.5000	0	.	0

New Patient Prediction

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Obs	_TYPE_	_FREQ_	min1	min2	max1	max2	avg_fre	max_fre	avgfre_bin	WOE
1	0	195	88.333	102.145	260.105	592.03	130	200	Medium	0.2

Obs	norm_avg	norm_max	I_status	P_1	P_0
1	0.24257	0.19975	1	0.75142	0.24858